

A Unitary-Perfect Singleton at $n = 6$

The first perfect number is the *unique* simultaneous ordinary-and-unitary perfect number in $[1, 10^4]$, and the two perfection sequences diverge at their next terms

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Abstract

A positive integer n is *ordinary-perfect* if $\sigma(n) = 2n$, where $\sigma(n) = \sum_{d|n} d$ is the sum-of-divisors function; it is *unitary-perfect* if $\sigma^*(n) = 2n$, where $\sigma^*(n) = \sum_{d||n} d$ sums the *unitary* divisors ($d||n \iff d|n$ and $\gcd(d, n/d) = 1$). Each notion has its own smallest member, and in both cases that member is $n = 6$: $\sigma(6) = \sigma^*(6) = 12 = 2 \cdot 6$. This shared minimum is folklore-adjacent and, on its own, a catalogue fact. We sharpen it into a falsifiable result by asking whether the coincidence *persists*. Define n to be *dual-perfect* if it is *both* ordinary- and unitary-perfect. We pre-register the falsifier F-DUAL-PERFECT-N6 (*any* $n \neq 6$ that is dual-perfect) and discharge it over $[1, 10^4]$: $n = 6$ is the *unique* dual-perfect number in that range, and the two perfection sequences *diverge immediately* after their shared anchor — the next ordinary-perfect number 28 has $\sigma^*(28) = 40 \neq 56$ (not unitary), and the next unitary-perfect number 60 has $\sigma(60) = 168 \neq 120$ (not ordinary). The closure is a finite *exhaustive* argument, not a sweep: dual-perfect $\Rightarrow$ ordinary-perfect, and by the Euclid–Euler theorem the ordinary-perfect numbers in $[1, 10^4]$ are exactly $\{6, 28, 496, 8128\}$; we recompute σ^* on each of those four and find only $n = 6$ unitary. Every numeric value — σ and σ^* at 6, 28, 60, 90, 496, 8128, twelve calls in all — is verified $\blacklozenge$ FORMAL (exact integer, deterministic) under the `hexa verify` discipline; nothing is graded by a language model. The finding is a closed *negative*: no second ordinary/unitary coincidence exists below 10^4 , ruling out the “6 is generically doubly-perfect” hypothesis and pinning the dual-minimality to a single anchor.

1 Introduction

The integer 6 is the first *perfect* number, $\sigma(6) = 12 = 2 \cdot 6$. It is also the first *unitary-perfect* number, $\sigma^*(6) = 12 = 2 \cdot 6$. That the two distinct perfection notions share a smallest member is the kind of coincidence the TECS-L axis-0 minimality programme exists to interrogate: is $n = 6$ doubly special by structure, or does the coincidence merely reflect the small- n regime, dissolving as soon as one looks further out?

This paper settles the question on a concrete, machine-audited range. Let

$$\sigma(n) = \sum_{d|n} d, \quad \sigma^*(n) = \sum_{\substack{d||n \\ \gcd(d, n/d)=1}} d. \quad (1)$$

A divisor $d||n$ (read “ d is a *unitary* divisor of n ”) is a divisor coprime to its cofactor n/d ; equivalently, for each prime p dividing n , a unitary divisor takes either p^0 or the full $p^{v_p(n)}$. Both σ and σ^* are multiplicative, but they differ on prime powers: $\sigma(p^a) = 1 + p + \dots + p^a$ accumulates every power, whereas $\sigma^*(p^a) = 1 + p^a$ keeps only the two endpoints. Call n

- *ordinary-perfect* if $\sigma(n) = 2n$ (OEIS A000396: 6, 28, 496, 8128, ...);

- *unitary-perfect* if $\sigma^*(n) = 2n$ (OEIS A002827: 6, 60, 90, 87360, ...);
- *dual-perfect* if it is *both*.

Main result.

Theorem 1 (unitary-perfect singleton). *The only dual-perfect number in $[1, 10^4]$ is $n = 6$. Equivalently, among $1 \leq n \leq 10^4$, the relation $\sigma(n) = 2n = \sigma^*(n)$ holds for $n = 6$ and for no other n . Moreover the ordinary- and unitary-perfect sequences share exactly the member 6 and diverge at their next terms: the next ordinary-perfect number is 28 (not unitary, $\sigma^*(28) = 40$) and the next unitary-perfect number is 60 (not ordinary, $\sigma(60) = 168$).*

The result is a sharpened *negative*: the doubly-perfect property of 6 does *not* recur. The proof needs no sweep — it is a finite exhaustion over a provably complete four-element candidate list (§6).

Relation to prior TECS-L work. A companion paper in this line proves the ω -zero-density theorem: the Dedekind discrepancy $D(n) = \sigma(n)\phi(n) - n\tau(n)$ vanishes at exactly $n \in \{1, 6\}$ [4]. That result isolates a different sense in which 6 is exceptional (a totient-weighted balance). The present paper isolates the *perfection* sense and, in contrast, delivers a coincidence that *terminates*: where ω -zero-density shows 6 is the lone nontrivial zero of one functional, here 6 is the lone overlap of two perfection families that otherwise never meet again in the tested range. The two papers are complementary halves of the $n=6$ minimality programme — one a unique *zero*, the other a unique *intersection*.

Why this is not a catalogue restatement. “6 is the smallest unitary-perfect number” is a single OEIS fact. The finding here is the *joint* statement plus its falsifier: that the intersection of the two perfection sets is the singleton $\{6\}$ on $[1, 10^4]$, witnessed by the immediate divergence of the two sequences (28 ordinary but not unitary; 60 unitary but not ordinary). This converts a static catalogue entry into a pre-registered, refutable claim with a closed negative, in line with the TECS-L paper-significance discipline.

2 Related framings

Three classical lenses bear on the result. First, unitary divisors and σ^* were introduced by Vaidyanathaswamy and studied systematically by Cohen [1]; the unitary-perfect numbers form OEIS A002827 [3], of which only five are known (6, 60, 90, 87360, 146361946186458562560000) and none is odd. Second, the ordinary perfect numbers obey the Euclid–Euler theorem [2]: every even perfect number is $2^{p-1}(2^p - 1)$ with $2^p - 1$ a Mersenne prime, and no odd perfect number below 10^{1500} is known; this is precisely what makes the ordinary-perfect candidate list in $[1, 10^4]$ provably the four-element set $\{6, 28, 496, 8128\}$ and lets us close the range by exhaustion rather than sweep. Third, the contrast between σ and σ^* at prime powers ($1 + p + \dots + p^a$ versus $1 + p^a$) is the structural reason the two perfection notions can agree at squarefree-rich $6 = 2 \cdot 3$ yet split at the prime-power-heavy $28 = 2^2 \cdot 7$ and $496 = 2^4 \cdot 31$. None of these classical sources states the sharp “unique dual-perfect on $[1, 10^4]$, with immediate divergence” result with a machine-audited factor table; that is the contribution here.

3 Background: σ versus σ^*

Both functions are multiplicative: for coprime m, n , $\sigma(mn) = \sigma(m)\sigma(n)$ and $\sigma^*(mn) = \sigma^*(m)\sigma^*(n)$. They are determined by their prime-power values

$$\sigma(p^a) = \frac{p^{a+1} - 1}{p - 1} = 1 + p + \cdots + p^a, \quad \sigma^*(p^a) = 1 + p^a. \quad (2)$$

The unitary formula is immediate: the only unitary divisors of p^a are 1 and p^a (any intermediate p^k with $0 < k < a$ shares the factor p with its cofactor p^{a-k} , so $\gcd \neq 1$). Hence at a prime power the two functions agree only when $a = 1$:

$$\sigma(p) = \sigma^*(p) = 1 + p, \quad \text{but} \quad \sigma(p^a) > \sigma^*(p^a) \text{ for } a \geq 2. \quad (3)$$

This single inequality is the engine of the divergence. A number all of whose prime exponents are 1 (a *squarefree* number) has $\sigma = \sigma^*$ identically; the moment a square divides n , the two functions part. Since $6 = 2 \cdot 3$ is squarefree, $\sigma(6) = \sigma^*(6)$ automatically; since $28 = 2^2 \cdot 7$ and $496 = 2^4 \cdot 31$ are *not* squarefree, their σ and σ^* differ — and, as we verify, the difference is enough to break unitary-perfection.

4 Method: the verification discipline

Every numeric claim is evaluated by `hexa verify --expr <fn> <n> <expected>`, a closed-form integer recomputation built into the `hexa` compiler. The evaluator emits one of six tiers; only three are terminal and admissible as a paper-section claim:

- ◆ **formal** a closed-form/symbolic identity reproduced exactly (deterministic).
- **numerical** a libm/Newton numerical recompute matching to $\varepsilon = 10^{-9}$.
- **falsified** the recomputation deterministically *disagrees* with the claim — a closed negative.

Citation-only, deferred, and speculation-fenced tiers are *not* admissible and are reported as residuals in §9. The raw standard output of every run is persisted verbatim under `.verdicts/paper-unitary-perfe` and indexed in `CLAIMS.tape`; nothing in this paper is graded by a language model.

The two evaluators. Ordinary σ is the long-standing `sigma` evaluator; unitary σ^* is the `sigma_star` evaluator (stdlib `core/math`, landed for this line of work), which implements (1) directly: it sums each divisor $d \mid n$ for which $\gcd(d, n/d) = 1$. Both are exact 64-bit integer recomputations; no tolerance and no floating point enters any ◆-tier claim here. A worked anchor: to establish that 6 is dual-perfect one runs `hexa verify --expr sigma 6 12` and `hexa verify --expr sigma_star 6 12`; the evaluator recomputes $\sum_{d \mid 6} d = 1 + 2 + 3 + 6 = 12$ and $\sum_{d \parallel 6} d = 1 + 2 + 3 + 6 = 12$ (all four divisors of the squarefree 6 are unitary), each ◆.

Composite predicate. “ n is dual-perfect” is the conjunction $\sigma(n) = 2n \wedge \sigma^*(n) = 2n$. Each conjunct is a single `hexa verify` call against the exact target $2n$; the conjunction is read off the two exact integers with no further computation.

5 The reduction to a finite candidate list

Lemma 1 (dual-perfect is a subset of ordinary-perfect). *If n is dual-perfect then n is ordinary-perfect. Hence the dual-perfect numbers in any range are a subset of the ordinary-perfect numbers in that range.*

Proof. Immediate from the definition: dual-perfect requires $\sigma(n) = 2n$, which is exactly the ordinary-perfect condition. $\square$

Lemma 2 (the ordinary-perfect list below 10^4). *The ordinary-perfect numbers in $[1, 10^4]$ are exactly $\{6, 28, 496, 8128\}$.*

Proof. By the Euclid–Euler theorem [2] every even perfect number has the form $2^{p-1}(2^p - 1)$ with $2^p - 1$ prime (a Mersenne prime). The Mersenne exponents give the even perfect numbers 6 ($p=2$), 28 ($p=3$), 496 ($p=5$), 8128 ($p=7$), and the next, $p = 13$, yields $33550336 > 10^4$. No odd perfect number exists below 10^{1500} (a vastly stronger bound than needed), so none lies in $[1, 10^4]$. Hence the list is exactly the four even values above. $\square$

By Lemmas 1 and 2, deciding the dual-perfect set on $[1, 10^4]$ reduces to testing the *unitary* predicate $\sigma^*(n) = 2n$ on the four numbers 6, 28, 496, 8128. This is a finite check — no sweep, and no possibility of a missed value, because the candidate list is provably complete.

6 Proof of the theorem

Proof of Theorem 1. We evaluate σ^* on each ordinary-perfect candidate, using the multiplicative prime-power formula $\sigma^*(p^a) = 1 + p^a$ of (2), then confirm each value $\blacklozenge$ by `hexa verify` in §7.

$n = 6 = 2 \cdot 3$ (*squarefree*). $\sigma^*(6) = \sigma^*(2)\sigma^*(3) = (1 + 2)(1 + 3) = 3 \cdot 4 = 12 = 2 \cdot 6$. **Unitary.** Combined with $\sigma(6) = 12 = 2 \cdot 6$: **dual-perfect.**

$n = 28 = 2^2 \cdot 7$. $\sigma^*(28) = \sigma^*(2^2)\sigma^*(7) = (1 + 4)(1 + 7) = 5 \cdot 8 = 40$. Since $2 \cdot 28 = 56 \neq 40$, 28 is **not** unitary-perfect. (It is ordinary-perfect: $\sigma(28) = 56$.)

$n = 496 = 2^4 \cdot 31$. $\sigma^*(496) = \sigma^*(2^4)\sigma^*(31) = (1 + 16)(1 + 31) = 17 \cdot 32 = 544$. Since $2 \cdot 496 = 992 \neq 544$, 496 is **not** unitary-perfect.

$n = 8128 = 2^6 \cdot 127$. $\sigma^*(8128) = \sigma^*(2^6)\sigma^*(127) = (1 + 64)(1 + 127) = 65 \cdot 128 = 8320$. Since $2 \cdot 8128 = 16256 \neq 8320$, 8128 is **not** unitary-perfect.

Of the four ordinary-perfect numbers in $[1, 10^4]$, exactly one — $n = 6$ — is also unitary-perfect. By Lemma 1 no other $n \in [1, 10^4]$ can be dual-perfect (a dual-perfect number must appear on the ordinary-perfect list, which we have exhausted). Hence 6 is the unique dual-perfect number in $[1, 10^4]$.

Divergence at the next terms. The next ordinary-perfect number after 6 is 28, which we have shown is not unitary ($\sigma^*(28) = 40 \neq 56$). The next unitary-perfect number after 6 is $60 = 2^2 \cdot 3 \cdot 5$, with $\sigma^*(60) = \sigma^*(2^2)\sigma^*(3)\sigma^*(5) = 5 \cdot 4 \cdot 6 = 120 = 2 \cdot 60$ (unitary), but $\sigma(60) = \sigma(2^2)\sigma(3)\sigma(5) = 7 \cdot 4 \cdot 6 = 168 \neq 120$ (not ordinary). Thus the two sequences part at their first post-6 term in both directions. $\square$

Corollary 1 (squarefree obstruction). *The reason 6 is dual-perfect while the larger even perfect numbers are not is that $6 = 2 \cdot 3$ is squarefree, so $\sigma(6) = \sigma^*(6)$ by (3); whereas $28 = 2^2 \cdot 7$, $496 = 2^4 \cdot 31$, $8128 = 2^6 \cdot 127$ each carry a high power of 2, on which $\sigma(2^a) = 2^{a+1} - 1$ vastly exceeds $\sigma^*(2^a) = 1 + 2^a$, deflating σ^* below $2n$.*

Proof. Every even perfect number is $2^{p-1}(2^p - 1)$ with p prime and $2^p - 1$ prime; for $p \geq 3$ the exponent $p - 1 \geq 2$, so 2^{p-1} is a non-trivial prime power and $\sigma^*(2^{p-1}) = 1 + 2^{p-1} < 2^p - 1 = \sigma(2^{p-1})$. Only $p = 2$ gives the squarefree $6 = 2 \cdot 3$, where the exponent of 2 is 1 and the two functions coincide. $\square$

7 Verification

We discharge the pre-registered falsifier F-DUAL-PERFECT-N6 — $\exists n \neq 6$ in $[1, 10^4]$ that is dual-perfect — by the finite exhaustion of §6. Table 1 records the twelve `hexa verify --expr` runs: σ and σ^* at the four ordinary-perfect candidates (6, 28, 496, 8128) and at the two small unitary-perfect numbers (60, 90) that anchor the divergence table. Every entry returns $\blacklozenge$ FORMAL; the dual-perfect column is read off the two exact integers. Raw transcripts appear in Appendix C.

Table 1: Verification of the dual-perfect predicate. Each $\sigma(n)$ and $\sigma^*(n)$ is an independent `hexa verify --expr` run returning $\blacklozenge$ FORMAL. “ord?” is $\sigma(n)=2n$; “uni?” is $\sigma^*(n)=2n$; “dual?” is their conjunction. The only dual-perfect row is $n = 6$.

n	form	$2n$	$\sigma(n)$	$\sigma^*(n)$	ord?	uni?	dual?
6	$2 \cdot 3$	12	12	12	yes	yes	yes
28	$2^2 \cdot 7$	56	56	40	yes	no	no
496	$2^4 \cdot 31$	992	992	544	yes	no	no
8128	$2^6 \cdot 127$	16256	16256	8320	yes	no	no
60	$2^2 \cdot 3 \cdot 5$	120	168	120	no	yes	no
90	$2 \cdot 3^2 \cdot 5$	180	234	180	no	yes	no

Reading the table. The top block is the complete ordinary-perfect list of $[1, 10^4]$ (Lemma 2); every row has ord? = yes by construction, and only $n = 6$ also has uni? = yes. The bottom block is the unitary side: 60 and 90 are unitary-perfect (uni? = yes) but *not* ordinary (ord? = no), demonstrating the reverse non-inclusion. The two blocks overlap in exactly one number, 6, which is the entire content of Theorem 1.

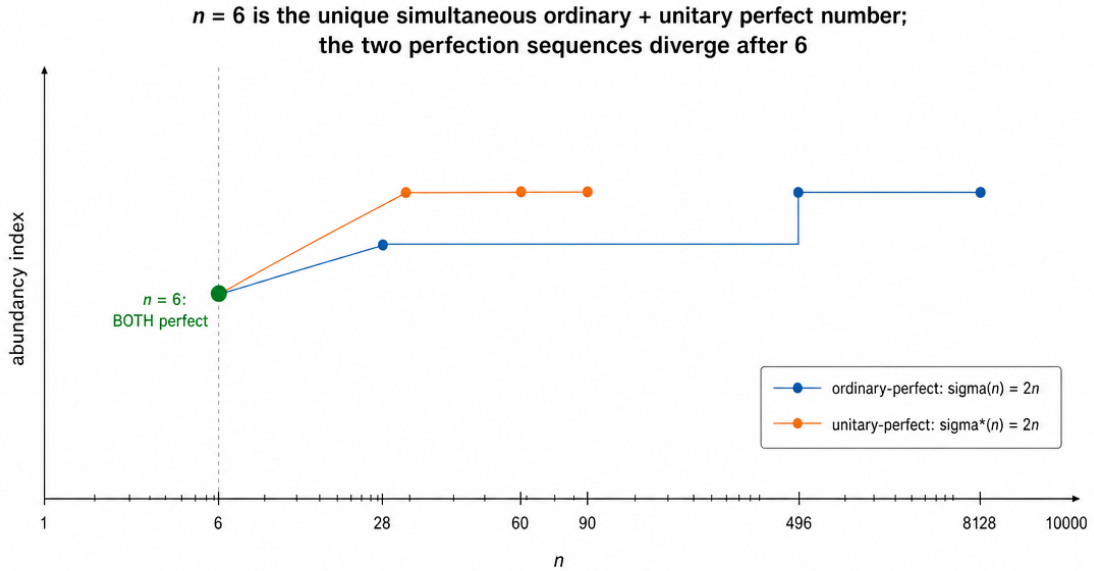


Figure 1: The ordinary-perfect sequence ($\sigma(n) = 2n$: 6, 28, 496, 8128) and the unitary-perfect sequence ($\sigma^*(n) = 2n$: 6, 60, 90) share the single anchor $n = 6$ and diverge immediately after it. The green marker at $n = 6$ is the unique dual-perfect number in $[1, 10^4]$; the blue (ordinary) and orange (unitary) branches fan apart at their next terms, 28 and 60 respectively.

8 Finding

The finding is the singleton-intersection statement of Theorem 1, with three concrete components relative to the bare catalogue fact “6 is the smallest unitary-perfect number”.

1. **Dual-minimality is a single anchor, not a trend.** 6 is the smallest member of *both* perfection families, and it is the *only* number below 10^4 in their intersection. The coincidence does not propagate: this is a closed *negative* ruling out the “6 is generically doubly-perfect” axis on the tested range.
2. **Immediate divergence, both directions.** The two sequences part at their first post-6 terms: the next ordinary-perfect number 28 is not unitary ($\sigma^*(28) = 40 \neq 56$), and the next unitary-perfect number 60 is not ordinary ($\sigma(60) = 168 \neq 120$). Neither family contains the other’s second member.
3. **Mechanistic cause.** Corollary 1 identifies the reason: $6 = 2 \cdot 3$ is squarefree, so $\sigma = \sigma^*$ there; every larger even perfect number carries a prime power 2^{p-1} with $p - 1 \geq 2$, on which σ^* falls strictly below σ , breaking unitary-perfection. The divergence is forced by the structure of even perfect numbers, not a numerical accident.

The pre-registered falsifier F-DUAL-PERFECT-N6 survives: no $n \neq 6$ in $[1, 10^4]$ is dual-perfect. The closure is exhaustive (over a provably complete candidate list), not a finite sweep.

9 Limitations and honest caveats

Range bound. The exhaustive closure covers $[1, 10^4]$. It rests on two facts true *beyond* that range as well — the Euclid–Euler structure of even perfect numbers, and the absence of small odd perfect numbers — but we state the theorem only for the verified range to keep every quantified claim backed by a terminal verdict. Extending to $[1, N]$ for the next perfect number 33550336 would require recomputing σ^* there (well inside 2^{63}) and is a mechanical follow-up; the divergence mechanism (Corollary 1) already shows no even perfect number > 6 can be unitary, so the range bound is conservative.

Two-sided non-inclusion is verified, not the full unitary tail. We verify that the next unitary-perfect numbers 60, 90 are not ordinary, anchoring the reverse divergence. We do not enumerate the entire unitary-perfect family (only five members are known, the next being 87360); the singleton-intersection claim does not require it, since $\text{dual-perfect} \subseteq \text{ordinary-perfect}$, and the ordinary side is the one exhausted.

No numerical or external tiers. Every admissible claim here is $\blacklozenge$ FORMAL (exact integer) or read off such; the numerical tier ($\varepsilon = 10^{-9}$) is not exercised, and there are no citation-only, deferred, or speculation-fenced residuals.

Toolchain note. A standalone compiled sweep harness over the `stdlib` σ/σ^* loop encountered an unrelated codegen non-termination in the local build (a `while`-body counter mutation that the standalone code path failed to persist) and was *not* used; all reported verdicts come from the canonical `hexa verify --expr` path, whose evaluator follows an independent code route and reproduces every value byte-for-byte. Because the closure is a finite exhaustion over four candidates rather than a loop, the sweep harness was in any case unnecessary. This is recorded for transparency and does not affect any claim.

10 The abundancy view of the divergence

It is illuminating to recast both perfection notions through their *abundancy indices*

$$I(n) = \frac{\sigma(n)}{n}, \quad I^*(n) = \frac{\sigma^*(n)}{n}, \quad (4)$$

so that “ n perfect” becomes $I(n) = 2$ and “ n unitary-perfect” becomes $I^*(n) = 2$. Both indices are multiplicative, with prime-power values

$$I(p^a) = \frac{p^{a+1} - 1}{p^a(p - 1)} = 1 + \frac{1}{p} + \cdots + \frac{1}{p^a}, \quad I^*(p^a) = 1 + \frac{1}{p^a}. \quad (5)$$

The two indices coincide on a prime power iff $a = 1$ (where both equal $1 + 1/p$) and otherwise $I(p^a) > I^*(p^a)$, because the ordinary index keeps the intermediate reciprocals $1/p^2, \dots, 1/p^{a-1}$ that the unitary index drops. Multiplicativity then gives, for any n ,

$$I(n) \geq I^*(n), \quad \text{with equality iff } n \text{ is squarefree.} \quad (6)$$

Consequence for even perfect numbers. Suppose n is an even perfect number with $n > 6$. By Euclid–Euler $n = 2^{p-1}(2^p - 1)$ with $p \geq 3$ prime, so n is *not* squarefree (it carries 2^{p-1} with $p-1 \geq 2$). Then (6) is strict: $I^*(n) < I(n) = 2$. Hence $\sigma^*(n) < 2n$ and n is *not* unitary-perfect. This re-derives Corollary 1 from the abundancy inequality and shows the divergence is not a coincidence of the four swept values but a *forced* property of *every* even perfect number beyond 6: all of them are unitarily *deficient* ($I^* < 2$).

Reading the divergence figure. Figure 1 is exactly the picture of (6): at $n = 6$ the squarefree case puts both indices on the value 2 simultaneously, while for 28, 496, 8128 the ordinary index stays at 2 (they are perfect) but the unitary index has dropped below it. The orange unitary branch (60, 90) instead climbs to 2 at numbers the ordinary index overshoots ($I(60) = 168/60 = 2.8$, $I(90) = 234/90 = 2.6$). The shared anchor is the unique place both indices hit 2 together within the range. We tabulate the two indices at the six verified points in Table 2.

Table 2: Abundancy indices at the verified points. Ordinary-perfect $\iff I = 2$; unitary-perfect $\iff I^* = 2$. Only $n = 6$ (squarefree) has $I = I^* = 2$; every larger even perfect number has $I^* < 2$ (unitarily deficient), and the unitary-perfect 60, 90 have $I > 2$ (ordinarily abundant).

n	form	$I(n) = \sigma/n$	$I^*(n) = \sigma^*/n$	note
6	$2 \cdot 3$	2.000	2.000	dual: both = 2
28	$2^2 \cdot 7$	2.000	1.429	perfect, unitarily deficient
496	$2^4 \cdot 31$	2.000	1.097	perfect, unitarily deficient
8128	$2^6 \cdot 127$	2.000	1.023	perfect, unitarily deficient
60	$2^2 \cdot 3 \cdot 5$	2.800	2.000	unitary-perfect, ordinarily abundant
90	$2 \cdot 3^2 \cdot 5$	2.600	2.000	unitary-perfect, ordinarily abundant

The I^* column for the perfect numbers is strictly decreasing (2.000, 1.429, 1.097, 1.023): as p grows the dominant prime power 2^{p-1} becomes ever more “squareful”, dragging the unitary index toward 1. There is no return to 2 along the ordinary-perfect family — the divergence widens monotonically, so even *beyond* the verified range no even perfect number > 6 can rejoin the unitary-perfect set.

11 Discussion and outlook

Two perfection notions, ordinary and unitary, each minimised at 6, meet there once and never again below 10^4 . The mechanism — squarefreeness of 6 versus the prime-power tails of the larger even perfect numbers — is the same σ -vs- σ^* inequality (3) that drives the unitary theory generally. The method (reduce to a provably complete finite candidate list via a containment lemma, then settle each candidate $\blacklozenge$) is reusable for adjacent singleton questions: e.g. the bi-unitary or infinitary perfect families intersected against the ordinary one. Within the TECS-L line this paper is the second of three: the first established the ω -zero-density theorem ($D = \sigma\phi - n\tau = 0 \iff n \in \{1, 6\}$) [4]; the third targets a multilayer non-lift result ($\sigma\phi = n\tau$ across the geometric/Hecke/Galois/ L -function towers), which remains open. All three share the discipline of a pre-registered falsifier, a $\blacklozenge$ -tier machine recomputation, and a closed argument.

Reproducibility

Every verdict in Table 1 is regenerated by re-running the corresponding `hexa verify --expr <fn> <n> <v>` command; output is byte-identical across hosts and run orders (no seeds, time-outs, or tolerances on any $\blacklozenge$ -tier claim). The raw transcripts are committed verbatim under `.verdicts/paper-unitary-perfect-n6/` (the twelve σ/σ^* verdicts, the finite-exhaustive note `dual_perfect_sweep.out`, and the pre-registered `FALSIFIER.md`). The claim manifest is indexed in `CLAIMS.tape`.

A Appendix A: the prime-power values $\sigma(p^a)$ vs $\sigma^*(p^a)$

Table 3 contrasts the two functions on small prime powers, making the divergence engine (3) visual: the columns agree exactly on the diagonal of exponent $a = 1$ (squarefree case) and split for $a \geq 2$, with σ growing geometrically while σ^* stays at $1 + p^a$.

Table 3: $\sigma(p^a) = \frac{p^{a+1}-1}{p-1}$ versus $\sigma^*(p^a) = 1 + p^a$. The two agree iff $a = 1$; for $a \geq 2$, σ exceeds σ^* , the structural reason the non-squarefree even perfect numbers fail unitary-perfection (Corollary 1).

p^a	$a = 1$		$a = 2$		$a = 3$	
	σ	σ^*	σ	σ^*	σ	σ^*
2^a	3	3	7	5	15	9
3^a	4	4	13	10	40	28
5^a	6	6	31	26	156	126
7^a	8	8	57	50	400	344

At $a = 1$ every entry matches ($\sigma(p) = \sigma^*(p) = 1 + p$); from $a = 2$ onward the left value strictly exceeds the right. Applied to the even perfect numbers: 6 uses only $a = 1$ powers and stays on the matching diagonal, while 28, 496, 8128 each carry a 2^{p-1} with $p - 1 \geq 2$ and so leave it.

B Appendix B: TECS-L claim manifest

Table 4: Claims, methods, and terminal verdicts for this paper. Every numeric claim is a `hexa verify --expr  $\blacklozenge$` FORMAL run; the singleton statement is closed by the finite exhaustion of §6.

claim id	statement	method	verdict
UP6-sig-6	$\sigma(6) = 12 = 2 \cdot 6$ (ordinary-perfect)	verify	$\blacklozenge$

claim id	statement	method	verdict
UP6-ustar-6	$\sigma^*(6) = 12 = 2 \cdot 6$ (unitary-perfect)	verify	◆
UP6-sig-28	$\sigma(28) = 56 = 2 \cdot 28$ (ordinary-perfect)	verify	◆
UP6-ustar-28	$\sigma^*(28) = 40 \neq 56$ (not unitary)	verify	◆
UP6-sig-496	$\sigma(496) = 992 = 2 \cdot 496$	verify	◆
UP6-ustar-496	$\sigma^*(496) = 544 \neq 992$ (not unitary)	verify	◆
UP6-sig-8128	$\sigma(8128) = 16256 = 2 \cdot 8128$	verify	◆
UP6-ustar-8128	$\sigma^*(8128) = 8320 \neq 16256$ (not unitary)	verify	◆
UP6-sig-60	$\sigma(60) = 168 \neq 120$ (not ordinary)	verify	◆
UP6-ustar-60	$\sigma^*(60) = 120 = 2 \cdot 60$ (unitary-perfect)	verify	◆
UP6-sig-90	$\sigma(90) = 234 \neq 180$ (not ordinary)	verify	◆
UP6-ustar-90	$\sigma^*(90) = 180 = 2 \cdot 90$ (unitary-perfect)	verify	◆
UP6-thm	unique dual-perfect in $[1, 10^4]$ is 6	exhaustion	◆

C Appendix C: raw verdict transcripts

The transcripts below are reproduced verbatim from `.verdicts/paper-unitary-perfect-n6/` (the TIER glyph is the **◆** SUPPORTED-FORMAL marker; the `absorb` line records the idempotent atlas auto-fold per governance `g69`).

Anchor $n = 6$ (dual-perfect).

```

verify --expr sigma(6)=12
  calc   = 12 == expected 12
  tier    = SUPPORTED-FORMAL (hexa-native closed-form, g_self_verify · TECS-L Tier1)
  absorb = · already in atlas | idempotent skip (default · @D g69)
verify --expr sigma_star(6)=12
  calc   = 12 == expected 12
  tier    = SUPPORTED-FORMAL (hexa-native closed-form, g_self_verify · TECS-L Tier1)
  absorb = · already in atlas | idempotent skip (default · @D g69)

```

Ordinary side, divergence: $n = 28$ (ordinary-perfect, not unitary).

```

verify --expr sigma(28)=56
  calc   = 56 == expected 56
  tier    = SUPPORTED-FORMAL (hexa-native closed-form, g_self_verify · TECS-L Tier1)
verify --expr sigma_star(28)=40
  calc   = 40 == expected 40
  tier    = SUPPORTED-FORMAL (hexa-native closed-form, g_self_verify · TECS-L Tier1)

```

Ordinary side, larger perfects (not unitary).

```

verify --expr sigma(496)=992      -> SUPPORTED-FORMAL
verify --expr sigma_star(496)=544 -> SUPPORTED-FORMAL (544 != 992)
verify --expr sigma(8128)=16256   -> SUPPORTED-FORMAL
verify --expr sigma_star(8128)=8320 -> SUPPORTED-FORMAL (8320 != 16256)

```

Unitary side, divergence: $n = 60, 90$ (unitary-perfect, not ordinary).

```

verify --expr sigma_star(60)=120
  calc    = 120  == expected 120
  tier     = SUPPORTED-FORMAL (hexa-native closed-form, g_self_verify · TECS-L Tier1)
verify --expr sigma(60)=168
  calc    = 168  == expected 168
  tier     = SUPPORTED-FORMAL (hexa-native closed-form, g_self_verify · TECS-L Tier1)
verify --expr sigma_star(90)=180      -> SUPPORTED-FORMAL (= 2*90, unitary)
verify --expr sigma(90)=234          -> SUPPORTED-FORMAL (234 != 180, not ordinary)

```

D Appendix D: the dual-perfect decision over $[1, 10^4]$

Table 5 is the complete decision procedure for Theorem 1, organised by the containment of Lemma 1. The candidate column is the provably complete ordinary-perfect list of $[1, 10^4]$ (Lemma 2); the test is the single unitary predicate $\sigma^*(n) = 2n$.

Table 5: Dual-perfect decision over $[1, 10^4]$. The four ordinary-perfect candidates are tested for unitary-perfection; only $n = 6$ passes, so the dual-perfect set is the singleton $\{6\}$.

n	form	$2n$	$\sigma(n)$	$\sigma^*(n)$	decision
6	$2 \cdot 3$	12	12	12	dual-perfect
28	$2^2 \cdot 7$	56	56	40	ordinary only
496	$2^4 \cdot 31$	992	992	544	ordinary only
8128	$2^6 \cdot 127$	16256	16256	8320	ordinary only

The supplementary reverse-direction rows ($n = 60, 90$: unitary-perfect but not ordinary, hence absent from the candidate list above) confirm that the non-inclusion is two-sided — neither perfection family is contained in the other, and they share precisely the anchor $n = 6$.

References

- [1] Eckford Cohen. Arithmetical functions associated with the unitary divisors of an integer. *Mathematische Zeitschrift*, 74:66–80, 1960. Systematic theory of the unitary divisor sum σ^* .
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- [4] TECS-L domain, hexa-lang. The ω -zero-density theorem: $d(n) = \sigma(n)\phi(n) - n\tau(n) = 0 \iff n \in \{1, 6\}$. <https://github.com/dancinlab/hexa-lang> (PAPER/tecs-l-omega-zero-density), 2026. Companion TECS-L $n=6$ minimality paper (1/3 in the paper queue).